Osptek Display

TFT LCD SPECIFICATION

Model No:

YDP146B001-V10

REVISION RECORD

REV NO. 版 本	CONTENTS 内 容	REVISION ITEM 修改项目	REV DATE 日 期
V1.0	First release	Preliminary	2023-05-06

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1. GENERAL SPECIFICATIONS(主要特征描述)

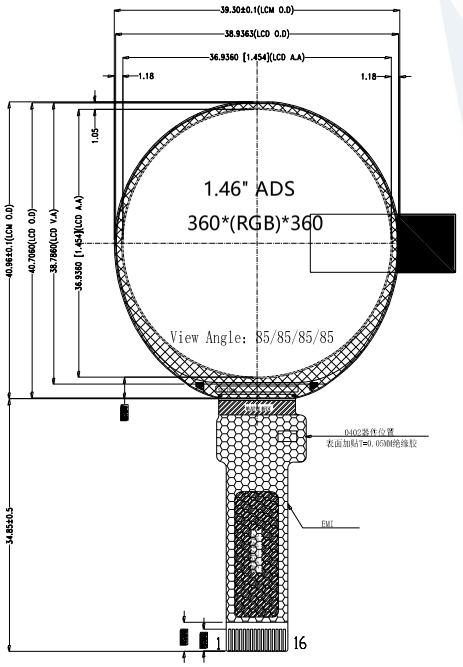
Item 项目	Contents 内容	Unit 单位
LCD type 液晶显示类型	262K TFT TRANSMISSVIE	/
Diagonal length 尺寸	1.46	inch
Recommended Viewing Direction 推荐使用方向	ALL	O'Clock
Dot arrangement 点阵	360*3(R.G.B)*360	Dot
Module size (W*H*T) 模块外围尺寸（宽*高*厚）	39.30*40.96*1.56	mm^3
Viewing area (W*H) 可视区域（宽*高）	38.93*38.78	mm^2
Active area (W*H) 有效显示区域（宽*高）	36.93*36.93	mm^2
Pixel size (W*H) 像素大小（宽*高）	0.1026*0.1026	mm^2
Backlight Type 背光类型	LED(white 3*LED)	/
Drive IC 驱动 IC	ST77916	/
Interface Type 接口类型	QSPI	/

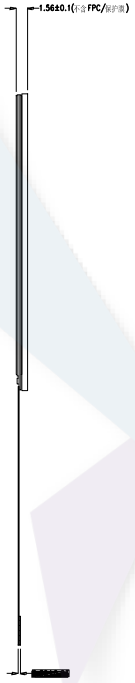
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2. EXTERNAL DIMENSIONS(外形尺寸)

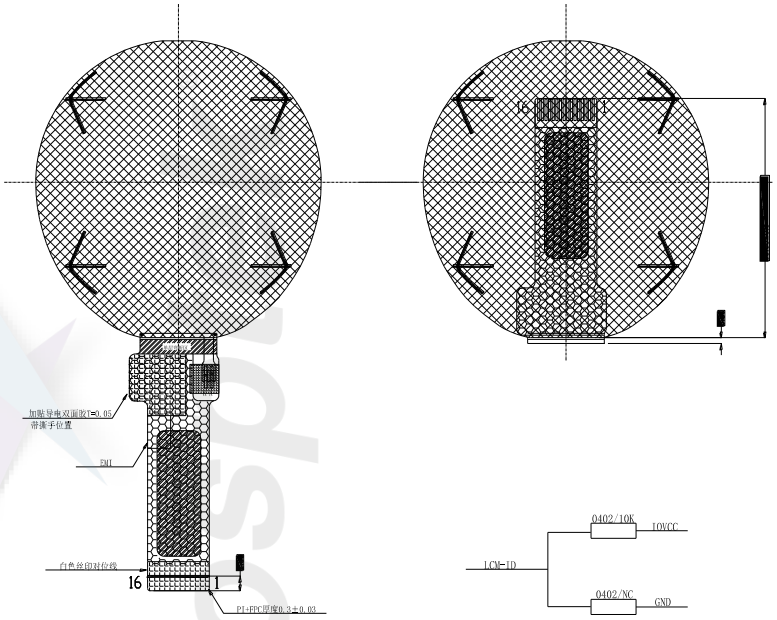
有害物质管理标准
■ 欧盟RoHS环保指令
□ 欧盟RoHS环保指令+无卤素要求
□ _____有害物质管理标准　第____版

REV	Date	Description
V01	20230424	NEW





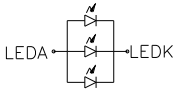
此四个位置要求在背光上
凹刻或丝印白色对位线。



Pin Assignment	
NO	SYMBOL
1	LEDK
2	LEDA
3	LCD_CS
4	GND
5	SPI_SIO0
6	SPI_SIO1
7	SPI_SIO2
8	SPI_SIO3
9	SPI_CLK
10	LCD_TE
11	LCD_RESET
12	VDD3.3V
13	VDD3.3V
14	GND
15	NC
16	LCD-ID

Specification:

1). Display mode: 1.52" 360*360 262K Colors IPS
2). Viewing angle: ALL
3). Operating temp.: -20°C~+60°C
Storage temp.: -20°C~+70°C
4). IC: ST77916
5). Backlight: 3 chip White LED
I LED=60mA　V LED=2.8V~3.4V
6). General Tolerance:±0.20.
7). Rohs Compliancy.
8). Recommended Case Open Area should be less than Module V.A;



If=60mA　Vf=2.8V~3.4V

背光灯电路图

FPC 弯折出货

Quad SPI

MATERIAL	FINISH	* CRITICAL DIMENSION / SPEC DIMENSION	深圳市鱼鹰光电科技有限公司 Shenzhen Osprey Optoelectronics Technology Co., Ltd.				
APPROVED			TITLE: LCM	SIZE: A4	UNIT: mm	SHEET NUMBER: 1 of 1	REVISION: 1
CHECKED			PRO NO.: YDP146B003-V10				
DESIGNED			DWG NO.:	3rd ANGLE	SCALE: 1:1		

3. ABSOLUTE MAXIMUM RATINGS(极限参数)

Parameter 参数	**Symbol** 符号	**Min** 最小值	**Max** 最大值	**Unit** 单位	**Remark** 备注
Operating temperature 操作温度	Top	-30	80	˚C	At 25±5˚C
Storage temperature 储存温度	Tst	-30	85	˚C	
Humidity 湿度	RH	/	90%(Max60˚C)	RH	
Power Voltage 电源电压	VDD	/	5.2	V	
	VDDIO	/	5.2	V	

4. ELECTRICAL CHARACTERISTICS(模块电气特性)

Parameter 参数	**Symbol** 符号	**Min** 最小值	**Typ** 典型值	**Max** 最大值	**Unit** 单位
Supply voltage for logic 逻辑电压	VDD	2.4	2.75	3.3	V
I/O power supply I/O 口电压	VDDIO	1.65	1.8	3.3	V
Input Current 输入电流	Nomal Mode IOVCC+VDD	-	7.5	8.5	mA
	Sleep Mode IOVCC+VDD	-	25	40	uA
Input voltage 'H' level 输入高电平	V_{IH}	0.7*VDDIO	-	VDDIO	V
Input voltage 'L' level 输入低电平	V_{IL}	0	-	0.3*VDDIO	V
Output voltage 'H' level 输出高电平	V_{OH}	0.8*VDDIO	-	VDDIO	V
Output voltage 'L' level 输出低电平	V_{OL}	0	-	0.2*VDDIO	V

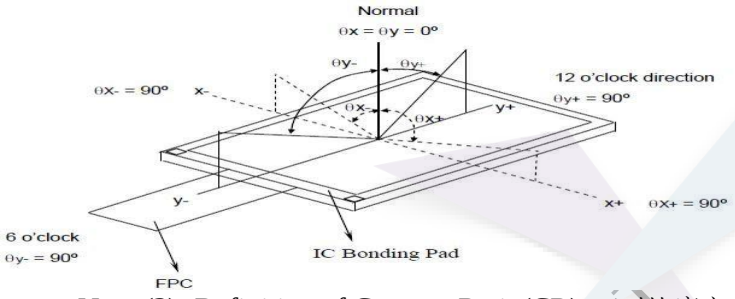
5. BACKLIGHT SPECIFICATION(背光电气特性)

Item 项目	**Symbol** 符号	**Min** 最小值	**Typ** 典型值	**Max** 最大值	**Unit** 单位	**Condition** 条件
Forward voltage 正向电压	V_f	2.8	3.0	3.3	V	I_f = 20 *3mA
Backlight uniformity 均匀度	No less than 80% （不低于 80%）					
Number of LED LED 数量	-	3			Piece	
Connection mode 连接模式	S/P/M	并联			-	

6. ELECTRO-OPTICAL CHARACTERISTICS(光电参数)

Item 项目		Symbol 符号	Condition 条件	Min 最小值	Typ 典型值	Max 最大值	Unit 单位	Note 备注
LCM Surface Luminance 模组表面亮度		Lv	--		500		cd/m^2	Note 4
Contrast Ratio 对比度		CR	--	900	1200		--	Note 2
Response time 响应时间		Tr+Tf	--		30	35	msec	Note 3
Color gamut 色域		S(%)		65	70		%	
Color chromaticity (CIE 1931) 色彩饱和度	White	X	θ=0° Ø=0° Ta=25°C		TBD			Note 4 Note 5
		Y			TBD			
	Red	X			TBD			
		Y			TBD			
	Green	X			TBD			
		Y			TBD			
	Blue	X			TBD			
		Y			TBD			
Viewing angle 视角范围	Ver.	Y+	CR≧10	80	85		deg	Note 1
		Y-		80	85			
	Hor.	X-		80	85			
		X+		80	85			

Note (1): Definition of Viewing Angle （视角定义）



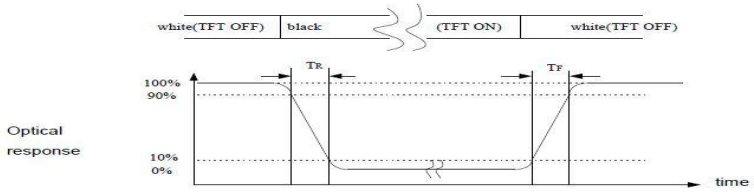
Note (2): Definition of Contrast Ratio(CR) （对比度定义）

Measured point 1 through 5 of panel.（测试 P1 到 P5 的亮度）

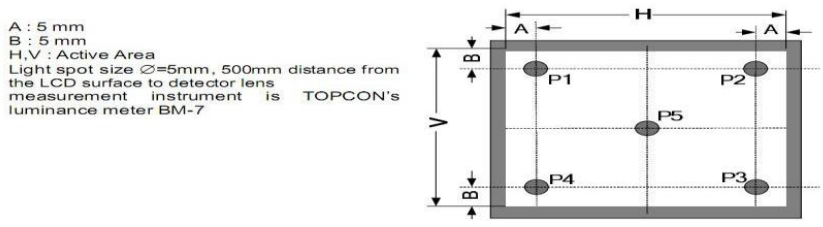
$$CR = \frac{\text{Luminance with all pixels white}}{\text{Luminance with all pixels black}}$$

$$\text{对比度} = \frac{\text{显示白色画面时平均表面亮度}}{\text{显示黑色画面时平均表面亮度}}$$

Note (3): Definition of Response Time: sum of TR and TF （响应时间定义）



Note (4): Measuring method for Contrast ratio, Surface Luminance, Luminance uniformity, CIE(x,y) Chromaticity （对比度，表面亮度，均匀度，CIE 坐标测试方法）



Note (5): CIE(x,y) Chromaticity, The X,Y value is determined by screen active area position 5. （CIE 坐标测试测试点为显示屏中心点P5）

7. INTERFACE DESCRIPTION(接口定义描述)

Pin No. Pin脚	Symbol 符 号	I/O 输入/出	Description 描 述	When not in use 未使用时的连接
1	LEDK	P	Power supply Cathode input for backlight	--
2	LEDA	P	Power supply Anode input for backlight	
3	CS	I	Chip selection pin	
4	GND	P	Ground	
5	SI00	IO	Serial Data input/output pin 0	
6	SI01	IO	Serial Data input/output pin 1	
7	SI02	IO	Serial Data input/output pin 2	
8	SI03	IO	Serial Data input/output pin 3	
9	LSCK	I	Serial Clock signal	
10	TE	I/O	Frame head pulse signal	
11	RESET	I	Reset signal pin	
12-13	VDD3.3V	P	Power supply to the internal Analog	
14	GND	P	Ground	
15	NC	/	NC	
16	LCD-ID	/	LCD-ID pin	

8. TIMING CHARACTERIST (时序特征)

8.1 AC Elerical Characteristics 模拟电气特性

Command write mode:

Write data for 16-bit/pixel(RGB 565 bit input), 65K Colors, 3AH='05h'

Write data for 18-bit/pixel(RGB 666 bit input), 262K Colors, 3AH='06h'

8.2 Reset timing 复位时间

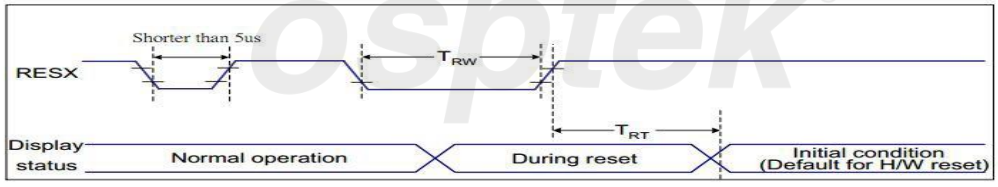


Figure 7 Reset Timing

VDDI=1.65 to 3.3V, VDD=2.4 to 3.3V, AGND=DGND=0V, Ta=-30 ~ 70 ℃

Related Pins	Symbol	Parameter	MIN	MAX	Unit
RESX	TRW	Reset pulse duration	10	-	us
	TRT	Reset cancel	-	5 (Note 1, 5)	ms
				120 (Note 1, 6, 7)	ms

Other information Reference IC datasheet. (ST77916)

9. ELIABILITY TEST CONDITIONS(可靠性实验条件)

NO. 序号	**Test Item** 实验项目	Test Condition 实验条件	Inspection after test 判定标准
1	High Temperature Storage 高温存放	70±2°C, 200 hours	Inspection after 2~4 hours storage at room tempearture, the sample

2	Low Temperature Storage 低温存放	-30±2°C,200 hours	shall be free from defects.(试验结束后须正常室温存放2~4个小时之后才能测试判定，不允许有以下缺陷) 1. Air bubble in the LCD（模块中有气泡）; 2. Seal leak（漏液）; 3. Non-display（不显示）; 4. Missing segments(漏笔); 5. Glass crack（玻璃破碎）; 6. Current IDD is twice higher than initial value（电流 Idd大于初时值的2倍）; 7. The surface damage（表面损伤） 8. Do not meet the electrical characteristics（不满足模块电气性能）
3	High Temperature Operating 高温操作	60±2°C,120 hours	
4	Low Temperature Operating 低温操作	-20±2°C,120 hours	
5	Damp proof Test Storage 高温高湿存放	40±2°C,90%RH,120 hours	
6	Temperature Cycle Storage 冷热循环存放	-20±2°C(30min) ~ 25°C(5min) ~ 70±2°C(30min), 10Cycle.	
7	Vibration Test 振荡试验	Frequency(频率):10HZ-55Hz , Amplitude（振幅）:1.5mm , x,y,z every direction for 1 hour (Packing condition) （包装状态，X,Y,Z 每个方向各 1 小时）	
8	Dropping Test 跌落试验	Drop to the ground from 1M height, one time, every side of carton (Packing condition) (包装状态，一米高度,6面各一次)	
9	ESD Test 静电测试	C:150pf; R:330Ω; Voltage:8KV; Air discharge(空气放电),10 times Voltage:4KV; Connet discharge(接触放电),10 times	

Remark（备注）:

1. The samples should be applied to only on test item(每个被测试样品只能用于其中的一个测试项目);
2. Sample size for each teat item is 5~10 pcs(每个测试项目的样品数量为 5~10 片);
3. For Damp Proof Test, Pure water(Resistance>10M Ω)should be used（对于防潮试验，试验箱的用 水必须是电阻大于 10M 欧姆的纯水）;
4. In case of malfunction defect caused by ESD damage, if it would be recovered to normal state after resetting ,it would be judged as a good part（如果由静电引起产品故障，当放置一段时间后能够恢复正常，则不视为产品缺陷）;
5. Failure Judgment Criterion: Basic Specification ,Electrical Characteristic, mechanical Characteristic, Optical Characteristic（故障判断标准：基本规格，电气特性，机械特性，光电特性）.

10.INSPECTION CRITERION(检查标准)(参考REX-BZ-001)

11.PRECAUTIONS FOR USING LCD MODULES(使用注意事项)

11.1 Using LCD Modules 使用

11.1.1 As glass is fragile, It tends to become or chipped during handling especially on the edges. Please avoid dropping or jarring. Do not subject it to a mechanical shock by dropping it or impact. (由于玻璃是脆的，使用过程请特别注意边缘区，防止跌落或振动，不能机械碰撞)。

11.1.2 Do not apply excessive force to the display surface or the adjoining are since this may cause the color tone to vary. Do not touch the display with bare hands. This will stain the display area and degraded insulation between terminals(some cosmetics are determined to the polarizer) (请勿施加过大的压力于显示屏或连接部位，否则会引起色调变化。不要裸手接触显示屏，这将弄脏显示区和降低端子之间的绝缘能力。一些外观问题是由偏光片决定的)。

11.1.3 The polarizer covering the display surface of the LCD module is soft and easily scratched. Handle this polarizer carefully. Do not touch, push or rub the exposed polarizes with anything harder than an HB pencil

lead (glass, tweezers, etc.). Do not put or attach anything on the display area to avoid leaving marks on it. Condensation on the surface and contact with terminals due to cold will damage, stain or dirty the polarizer. After products are tested at low temperature they must be warmed up in a container before coming in to contact with room temperature air.(覆盖液晶显示模块显示平面的偏光片是软性且易被擦伤，请小心轻拿。请勿用任何硬度大于 HB 铅笔芯的物品（玻璃，镊子等） 接触、撞压或摩擦裸露偏光片不要放置或粘附物体在显示区域上以免留下痕迹。冷凝在表面和端子将会损坏或弄脏偏光片。产品在 低温下测试之后，与室温空气接触之前必须在容器内升温)

11.1.4 If the display surface becomes contaminated, breathe on the surface and gently wipe it with a soft dry cloth. If it is heavily contaminated, moisten cloth with one of the following solvents: Isopropyl alcohol;Ethyl alcohol . Do not scrub hard to avoid damaging the display surface(如果显示平面受污，可对平面吹热气且轻轻地用软性干布擦除。如果受污严重，用含下列一种溶剂的湿布擦除:甘油，酒精 。请勿用力擦拭以免损坏显示平面。)

11.1.5 Solvents other than those above-mentioned may damage the polarizer. Especially, do not use the following: Water, Ketone, Aromatic solvents. Wipe off saliva or water drops immediately, contact with water over a long period of time may cause deformation or color fading. Avoid contact with oil and fats. (除以上提到的溶剂外，其他溶剂可能会损坏偏光片，特别要避免使用以下溶剂： 水，丙酮，芳烃溶剂。立即擦掉唾液或水滴，长时间与水接触会引起变形或褪色。避免接触油和油脂)

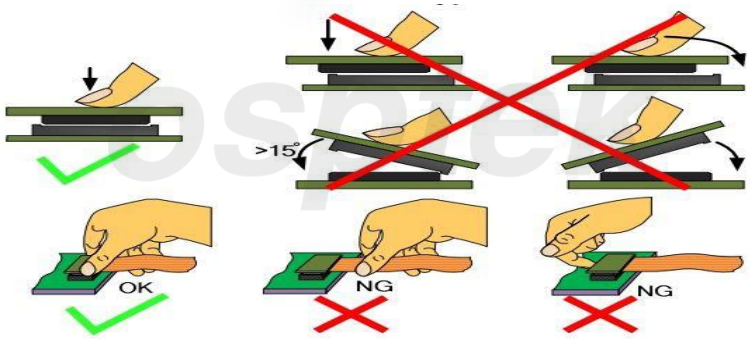
11.1.6 Do not attempt to disassemble or process the LCD module. （请勿拆卸液晶显示模块）

11.1.7 Electro-Static Discharge Control, Since this module uses a CMOS LSI, the same careful attention should be paid to electrostatic discharge as for an ordinary CMOS IC. To prevent destruction of the elements by static electricity, be careful to maintain an optimum work environment. （由于液晶显示模块使用 CMOS 集成，要特别注意静电放电问题。对 CMOS器件，要特别注意静电。为防止静电损坏，注意保持合宜的工作环境）

11.1.8 Input logic voltage before apply analog high voltage such as LCD driving voltage when power on. Remove analog high voltage before logic voltage when power off the module. Input each signal after the positive/negative voltage becomes stable. （开机时，先让逻辑电压，再接通模拟高压， 如显示屏驱动电压。关机时，先断开模拟高压，再关逻辑电压。正负电源都稳定后再送控制信号。）

11.1.9 In the use of connector products, the operating process of attention to turn off the power before pull off and insert action. To avoid damage to the module (在使用连接器的产品 时，插接过程注意先关闭电源再进行拔插动作，避免损坏模块)

11.1.10 Precaution for assemble the module with BTB connector:Please note the position of the male and female connector position, don’t assemble or assemble like the method which the following picture shows
（用板对板连接器安装液晶显示模块注意事项： 请注意连接器的公母及连接位置，请勿出现下图所示的错误连接方式）



11.2 Storage Modules 储存

11.2.1 Store them in a dark place. Do not expose to sunlight or fluorescent light, keep the temperature between 0° C and 35° C, and keep the relative humidity between 40%RH and 60%RH. （避光保存，避免直接暴露在太阳光或黄光灯下，保持温度在 0~35 摄氏度之间,保持相对湿度在 40%RH 和 60%RH 之间。）

11.2.2 The polarizer surface should not come in contact with any other objects (We advise you to store them in the anti-static electricity container in which they were shipped). （偏光片表 面避免接触其他物质，建议存放在货运防静电包装中）

11.3 Soldering

11.3.1 Iron head temperature（铬铁头实际温度）: 350±10℃, Soldering time（焊接时间）: <3-4S. Soldering don’t repeat above 3 times（焊接次数勿超过 3 次）

11.3.2 If soldering flux is used, be sure to remove any remaining flux after finishing to soldering operation (This does not apply in the of a non-halogen type of flux). It is recommended that you protect the LCD surface case with a cover during soldering to prevent any damage due to flux spatters. （如果使用助焊剂，完成焊接后一定要清除剩余的助焊剂（除非卤化物助焊剂）。建议焊接时用盖子保护显示屏面以避免因焊剂油溅出造成的任何损坏。）

12. **PRIOR CONSULT MATTER** (提前商议事项)

14.1 For OSP electronics standard products, we keep the right to change material, process ... for improving the product property without prior notice to our customer.
（对于**鱼鹰**電子的标准模块产品，我们保留在不通知客户的情况下,为提高产品性能而改变原材料及加工方法等的权利。 ）

14.2If you have special requirement about reliability condition, please let us know before you start the design on our samples.
（如对可靠性条件有特殊要求，请在模块设计开发前通知我们。）

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